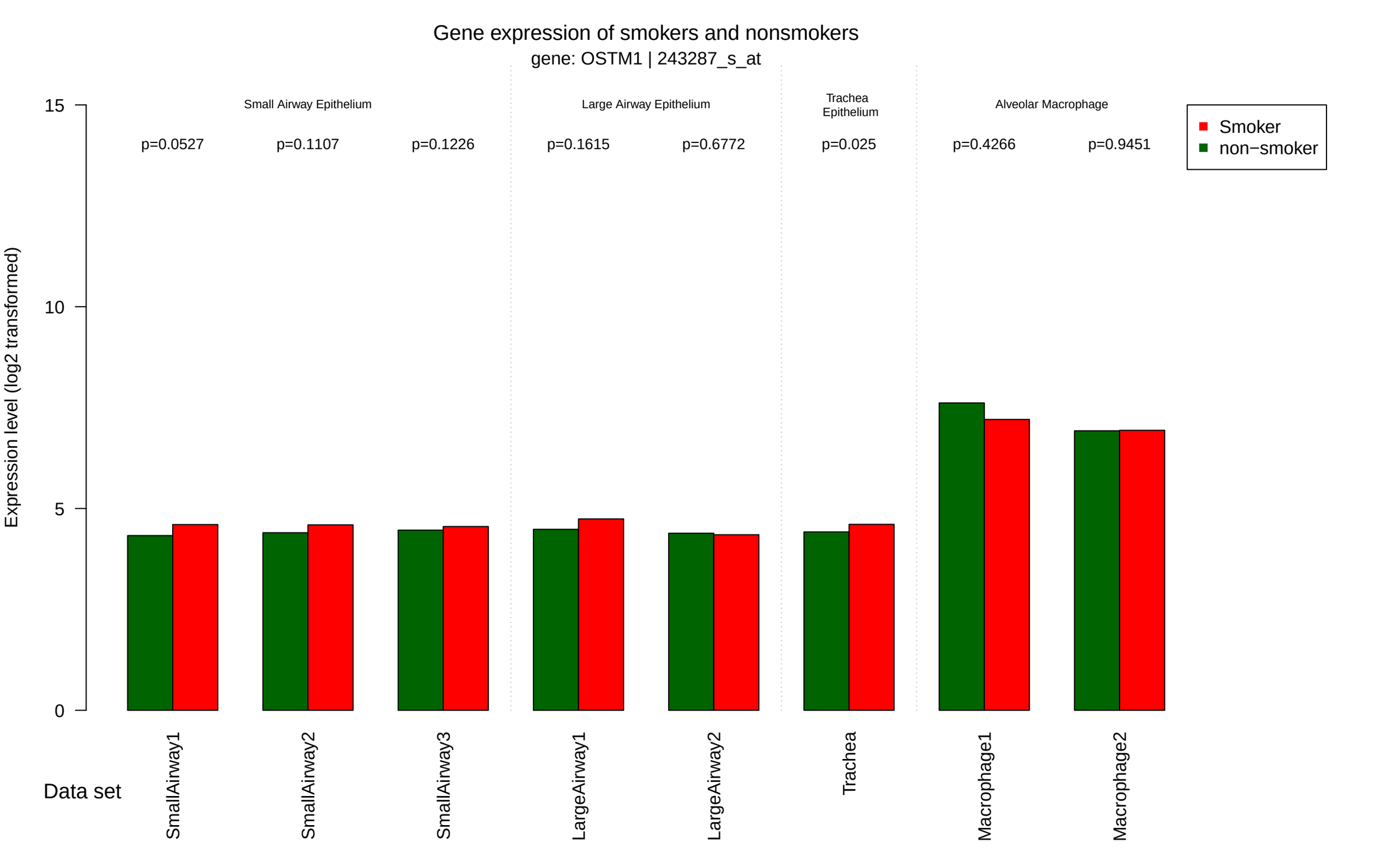
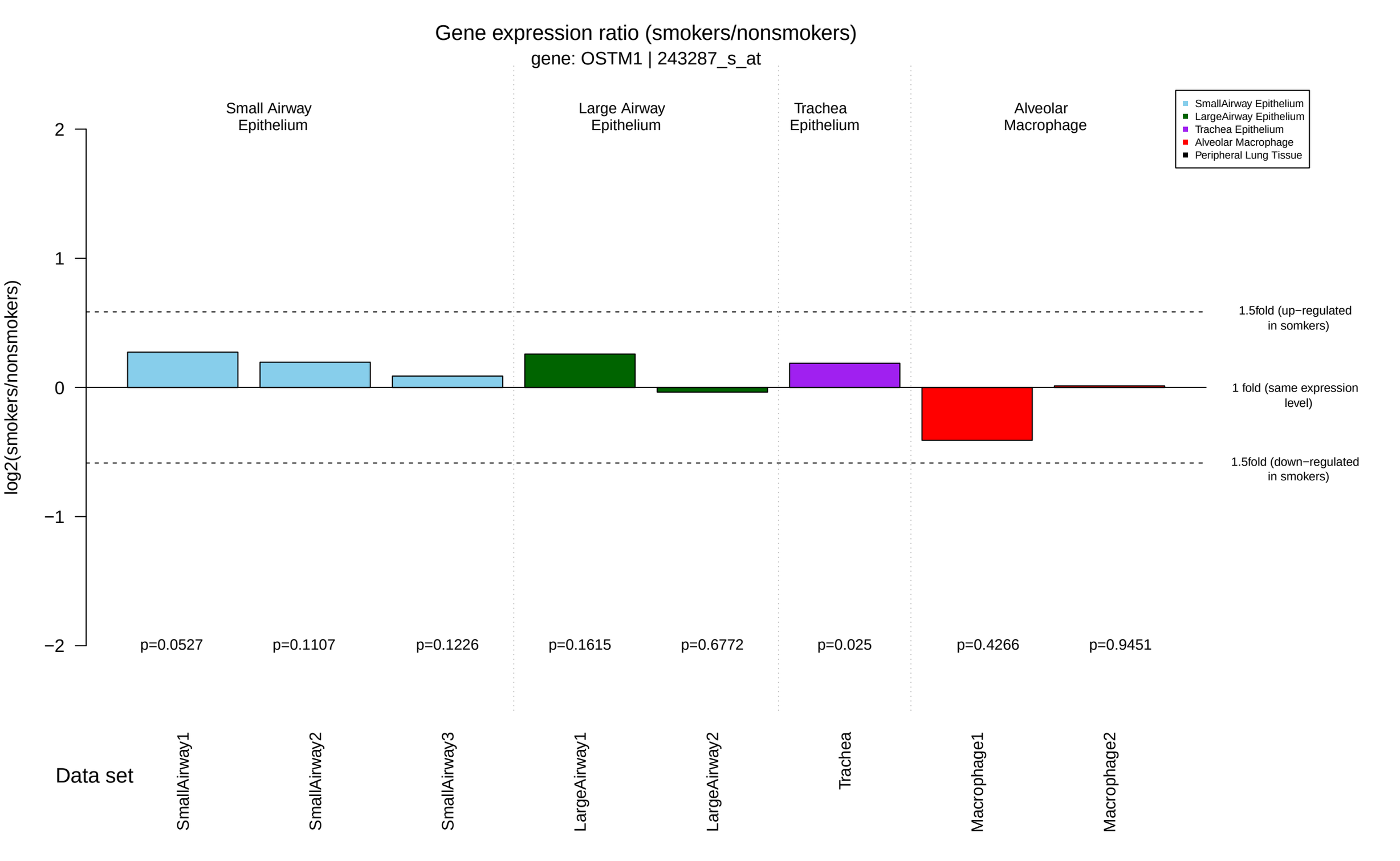
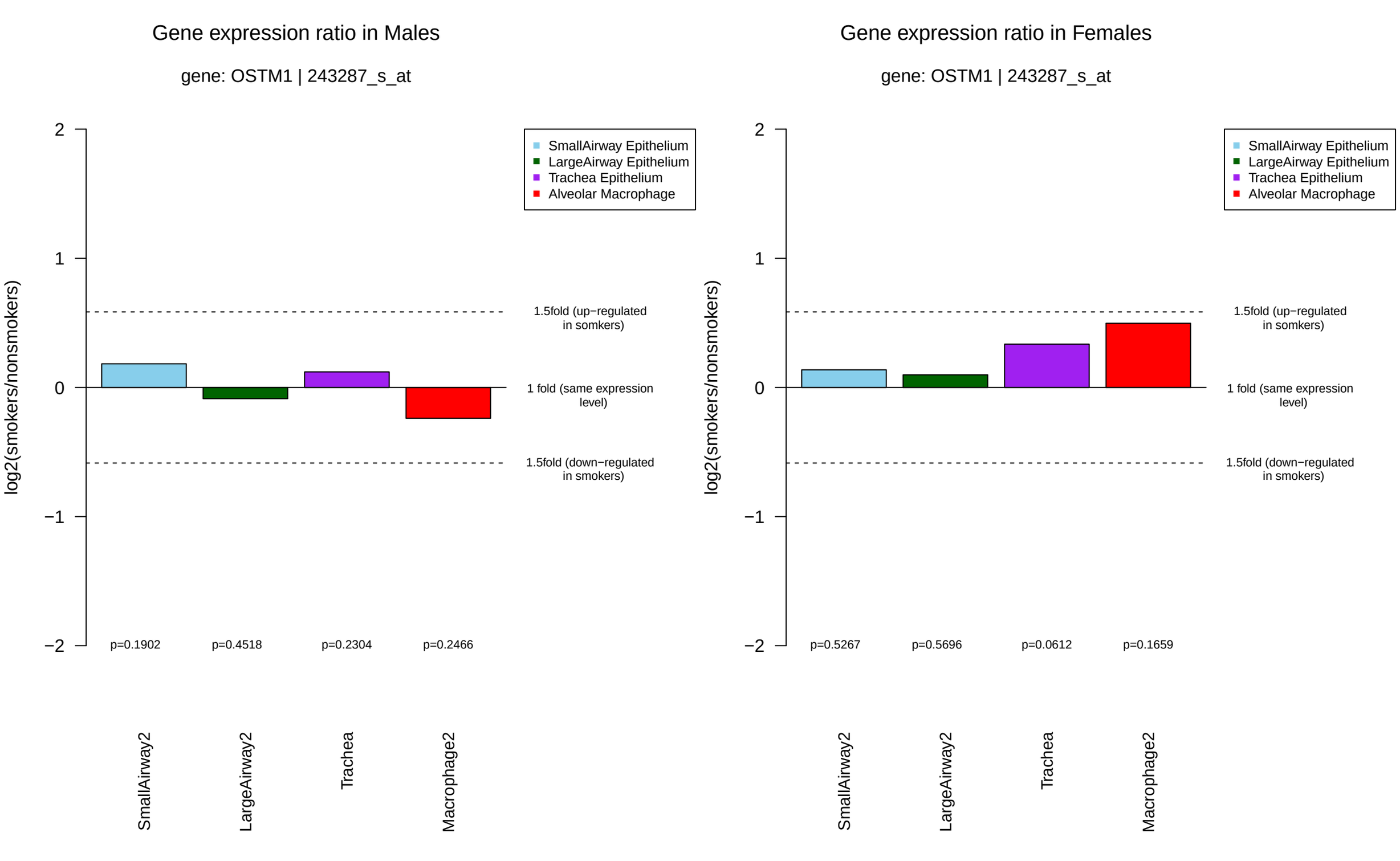
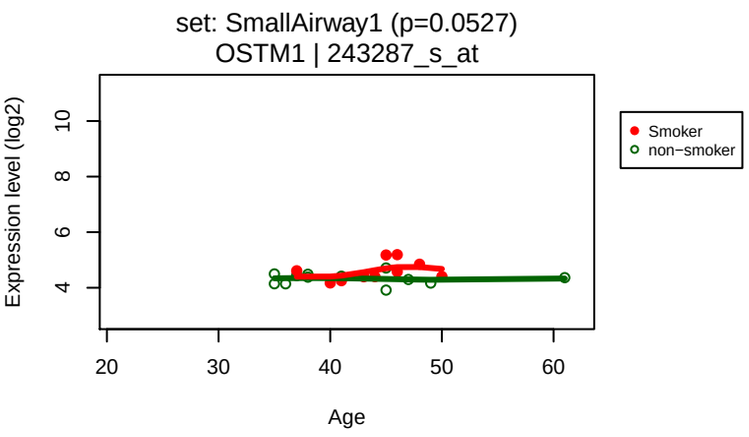
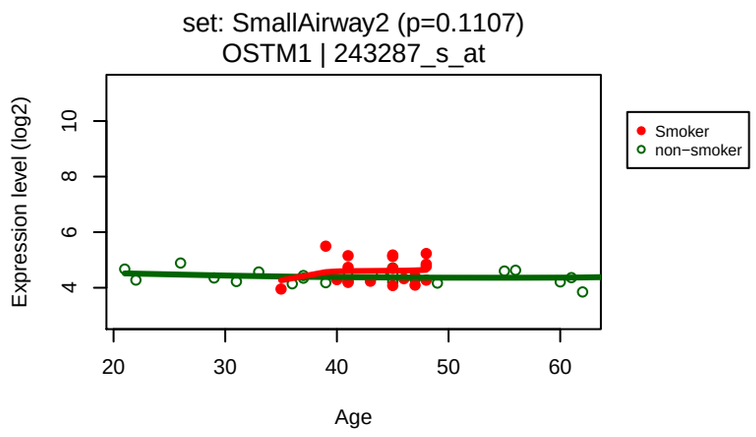
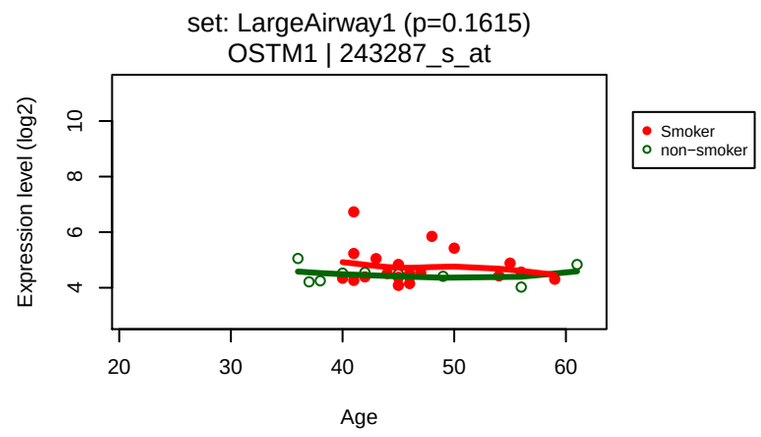

Gene expression of smokers and nonsmokers
gene: OSTM1 | 243287_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0527
p=0.1107
p=0.1226
p=0.1615
p=0.6772
p=0.025
p=0.4266
p=0.9451
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

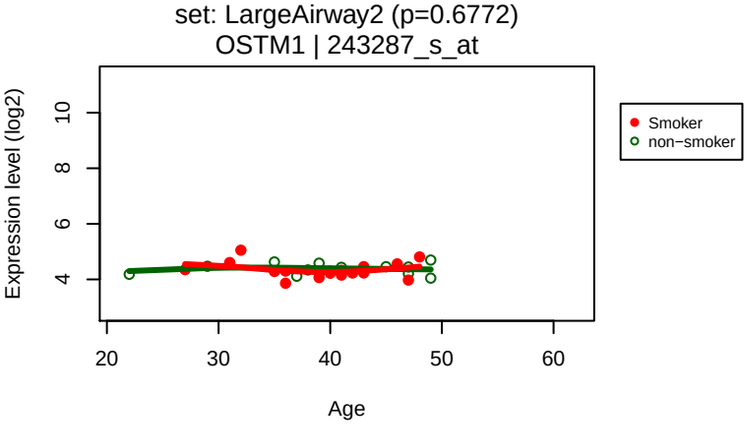

Gene expression ratio (smokers/nonsmokers)
gene: OSTM1 | 243287_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0527
p=0.1107
p=0.1226
p=0.1615
p=0.6772
p=0.025
p=0.4266
p=0.9451
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

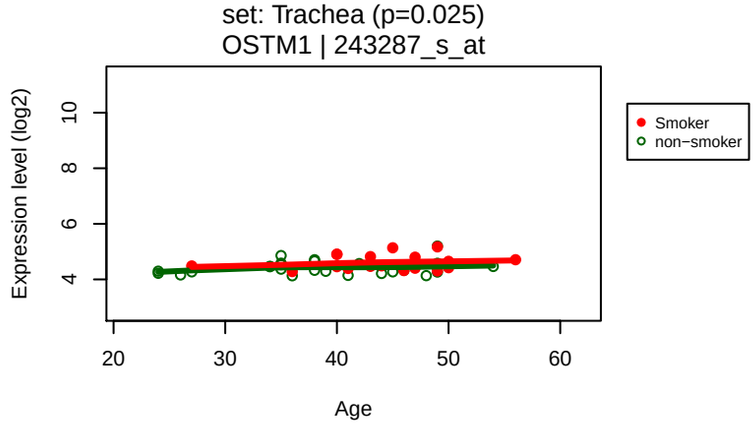

Gene expression ratio in Males
gene: OSTM1 | 243287_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1902
p=0.4518
p=0.2304
p=0.2466
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OSTM1 | 243287_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5267
p=0.5696
p=0.0612
p=0.1659
SmallAirway2
LargeAirway2
Trachea
Macrophage2

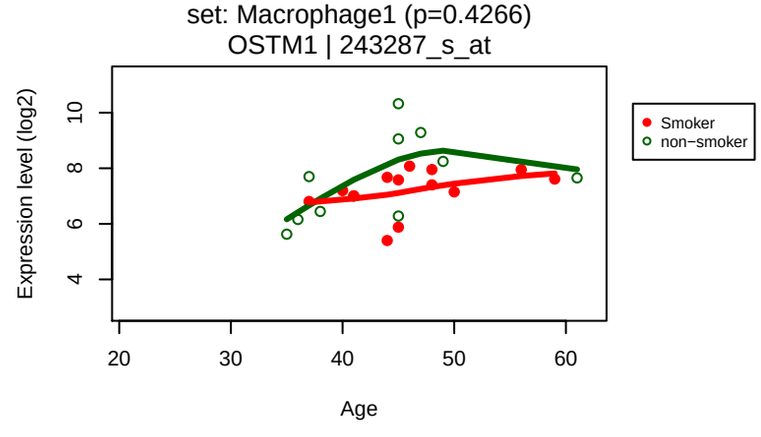

set: SmallAirway1 (p=0.0527)
OSTM1 | 243287_s_at
Expression level (log2)
Age
Smoker
non−smoker

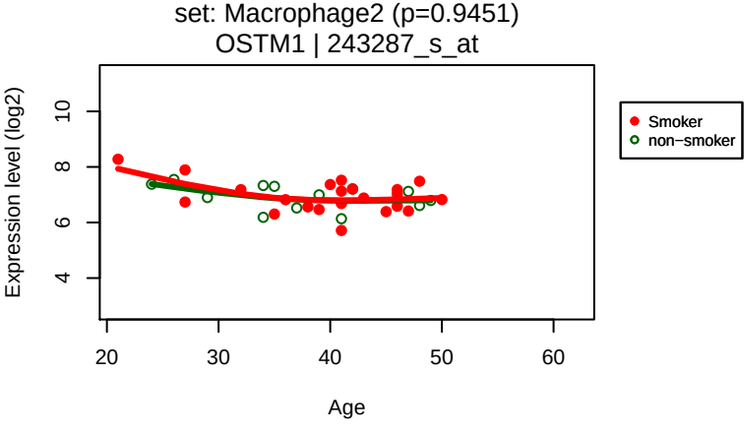

set: SmallAirway2 (p=0.1107)
OSTM1 | 243287_s_at
Expression level (log2)
Age
Smoker
non−smoker

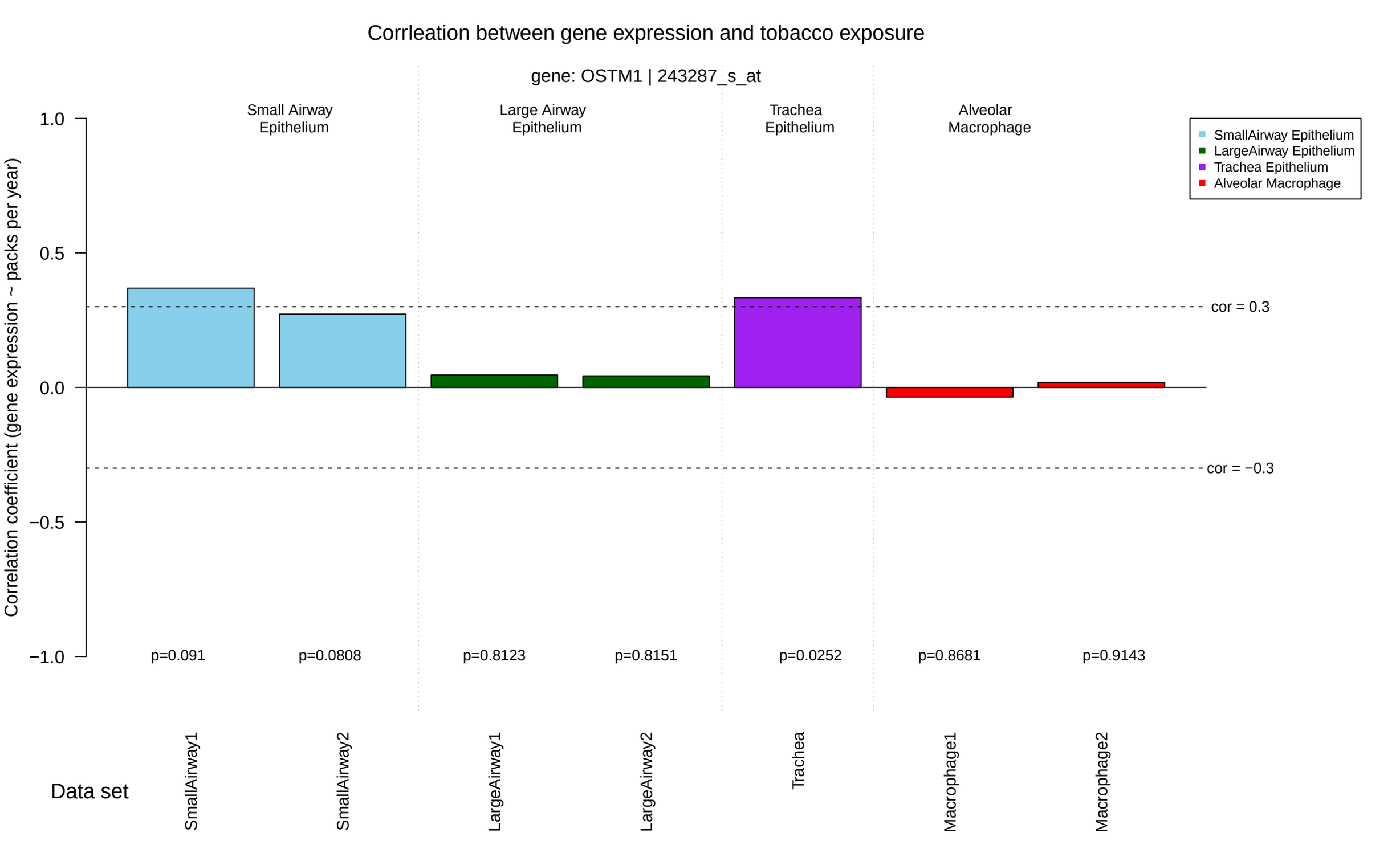

set: LargeAirway1 (p=0.1615)
OSTM1 | 243287_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6772)
OSTM1 | 243287_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.025)
OSTM1 | 243287_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4266)
OSTM1 | 243287_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9451)
OSTM1 | 243287_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OSTM1 | 243287_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.091
p=0.0808
p=0.8123
p=0.8151
p=0.0252
p=0.8681
p=0.9143
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2